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The reason we should get excited about the Cell processor is not just from the scope of gaming (which it will do immensely well as and when games are written for it, as it can handle and process multiple gaming instructions simultaneously, thus being able to give a more realistic appeal to the games) but also because the design of the Cell is extremely modular.

You can add more SPEs, or, in fact, remove some of them. Computing will become very modular in the future, and will give rise to a whole range of Cell-like processors depending on the kind of work load they're being targeted at.

Like the PPE, the SPEs have dual-issue cores, do not have a branch predictor, and rely completely on software for that function.

IBM calls the Cell a 'system on a chip' design, but it becomes a lot easier to understand it if you think of it as a network on a chip.

The SPEs

Each of the SPEs has 256 KB of local memory.

This is different from cache, as local memory can only be accessed by that SPE to which that local memory has been assigned.

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The SPEs are very basic processors, with the single-minded task of running through single- and double-precision mathematical calculations as fast as they can.

The SPEs are under direct control of the PPE, and hence they do not need to have their own personal instruction queuing logic (this is what decides what sequence the instructions are processed in) and hence are a lot smaller.

It is the PPE that does the work of scheduling, assigning and sending the data to each SPE for further processing.



The Xbox 360

The EIB

The EIB is a set of four 128-bit-wide data channels that connects the SPEs to the PPE, the Memory Interface controller as well as the Flex I/O system (which handles communication between the Cell chip and the rest of the computer system).

The Memory Interface Controller (MIC) connects to the EIB and handles the data transfer functions of the main memory. The approach taken here is the same as that of AMD, where the memory controller is built on the processor die, thus cutting out the need for data to be transferred from the processor to a separate memory controller—thereby leading to an overall boost in system performance.

The Flex I/O

The Flex I/O bus is like a regular front side bus (FSB) in a system and handles data transfer between the Cell processor and the rest of the system. It provides a whopping 76.8 GB/s data transfer rate, but this capacity is split into two parts—the data

that is intended for the Cell processor (or, in fact, other Cell processors in the system—termed as coherent data), and the data meant for the rest of the system (termed as incoherent data).

The RAM

Since the cell chip runs at 4 GHz and has massive memory requirements, the Cell is not based on the regular DDR (I, II or III), but on the RAMBUS technology. It has a dual-channel XDR memory bus running at 400 MHz.

Keep in mind two things: first, as has already been explained, on-die implementation greatly reduces memory latencies; second, XDR transfers data at eight times the memory bus clock, so you get 3.2 GHz of effective data signalling rates. While all this might be getting a bit too complex, suffice it to say that the end result of all this is a whopping 25.6 GB/s memory bandwidth!

The Unified Shader Architecture

To understand and appreciate the true merit of ATI's achievement in being able to develop this for the Xbox 360, let's first understand the basics behind shaders.

No gamer today can shy away from discussing pixel shaders and vertex shaders and their cumulative effect on the visual appeal of the game.

The pixel is the smallest element in an image or graphic. Pixel shaders let programmers control the characteristics of the appearance of the gaming environment on a per-pixel basis. That is, they allow them to make graphics look organic, instead of artificial and computer-generated.

Vertex shaders, on the other hand, provide the programmer with the ability to add special effects by creating highly realistic animations, such as deformation of a car's bonnet in a crash. Programmers use high-level languages such as the DirectX API to give us the highly realistic graphics we enjoy in games today.

With the Xbox 360, ATI has become the first manufacturer to provide Pixel and Vertex shader integration at a hardware level. This is different from the trend so far, where Pixel and Vertex shader integration was done at the software level using device drivers.

